# ConverseIQ Technical Overview & System Architecture

## 1. Executive Summary

ConverseIQ is an offline-first, voice-enabled document intelligence platform designed for privacy-preserving document retrieval and narration. It eliminates the need for cloud-based vector databases by leveraging a lightweight, high-precision hierarchical BM25 sparse retriever coupled with a cross-encoder reranker. The system runs efficiently on standard consumer CPU hardware, delivering high factual accuracy and low latency without transmitting sensitive user data to external cloud services.

## 2. Hierarchical Retrieval Architecture

Standard dense retrieval pipelines rely heavily on vector embeddings and high-dimensional indexes, which demand substantial memory and specialized hardware. In contrast, ConverseIQ adopts a three-tiered hierarchical chunking strategy: Section Level, Paragraph Level, and Sentence Level. During document ingestion, text extracted from PDFs is split dynamically along semantic section boundaries. When a user submits a query, BM25 operates across candidate paragraphs to rapidly narrow down the search space to the top 10 most relevant text blocks.

Following sparse candidate retrieval, a cross-encoder reranker based on MiniLM evaluates the top 10 paragraph candidates. The cross-encoder performs deep cross-attention between the query and candidate passages to compute precise semantic similarity scores. The top 3 reranked chunks are subsequently formatted into a structured context window and forwarded to the local language model for answer synthesis.

## 3. Local LLM Integration and Memory Management

ConverseIQ integrates with Ollama to run local open-weights language models such as Llama 3 and Gemma 2. To guarantee strict factual grounding and eliminate hallucinations, system prompts enforce a strict constraint: responses must be derived exclusively from the provided document context. If a question cannot be answered from the document chunks, the model explicitly responds that the information is unavailable in the document.

Conversational state is maintained using a sliding window memory mechanism with a configurable history window k=5. This allows users to ask multi-turn follow-up questions while preventing prompt overflow and managing local context limits efficiently.

## 4. System Performance and Resource Requirements

The average CPU latency for BM25 retrieval across a 50-page document is under 15 milliseconds. Cross-encoder reranking adds approximately 150 to 300 milliseconds on standard x86 and ARM processors. Local LLM token generation streams at 15 to 25 tokens per second depending on processor threads, ensuring an interactive and responsive chat experience.